

PERSONALIZED LINUX + GIT CHEAT SHEET

Online Activity #2 • Jay Fronda • A4 Quick Reference

BASIC LINUX COMMANDS	
pwd	Example: pwd Shows the full path of my current directory.
ls	Example: ls -la Lists files; -l gives details, -a includes hidden files. Flags: -l = long format, -a = all
cd	Example: cd ~/Documents Moves me to another directory. Flags: .. = parent, ~ = home
mkdir	Example: mkdir -p projects/web Creates a directory; -p also creates missing parent folders. Flags: -p
touch	Example: touch notes.txt Creates an empty file or updates its timestamp.
cp	Example: cp report.txt backup/ Copies files/directories. Flags: -r = recursive, -i = ask before overwrite
mv	Example: mv old.txt new.txt Moves a file or renames it. Flags: -i = ask before overwrite
rm	Example: rm -i notes.txt Deletes files; I use -i as a safety check. Flags: -i = prompt, -r = recursive
cat	Example: cat README.md Prints a file's contents directly in the terminal. Flags: -n = number lines
grep	Example: grep -n "error" log.txt Searches text for a matching pattern. Flags: -n = line number, -i = ignore case, -r = recursive
head / tail	Example: head -n 5 file.txt Shows the beginning / end of a file. Flags: -n = number of lines, -f = follow tail
chmod	Example: chmod +x script.sh Changes file permissions, such as making a script executable. Flags: +x = add execute permission

BASIC GIT COMMANDS	
git init	Example: git init my-project Starts a new Git repository in a directory. Flags: -b = initial branch, -q = quiet
git restore	Example: git restore --staged file.txt Restores files; commonly used to unstage or discard working-tree edits. Flags: --staged = unstage, --source= = choose source
git status	Example: git status -sb Summarizes what is changed, staged, or untracked. Flags: -s = short, -b = branch info
git branch	Example: git branch feature-login Creates, lists, renames, or deletes branches. Flags: -a = all, -d = delete, -m = rename
git add	Example: git add . Puts selected changes into the staging area for the next commit. Flags: . = current tree, -A = all changes
git switch / checkout	Example: git switch -c feature Changes branches; switch is the clearer modern command. Flags: -c = create + switch; checkout also has legacy file-restore use
git commit	Example: git commit -m "Fix login bug" Records the staged snapshot in repository history. Flags: -m = message, -a = stage tracked edits + commit
git merge	Example: git merge feature Combines another branch into the current branch. Flags: --no-ff = force merge commit, --abort = cancel conflicted merge
git log	Example: git log --oneline --graph Shows commit history so I can trace project changes. Flags: --oneline = compact, --graph = branch graph, -p = patch
git tag	Example: git tag v1.0.0 Names a specific commit, often for releases. Flags: -a = annotated tag, -d = delete
git diff	Example: git diff --staged Compares changes before I commit them. Flags: --staged = index vs HEAD, --stat = summary
git clone	Example: git clone Copies an existing remote repository to my computer. Flags: -b = clone a branch, --depth 1 = shallow clone

3-TREE ARCHITECTURE: HOW A CHANGE BECOMES A COMMIT		
1. WORKING DIRECTORY	2. STAGING AREA (INDEX)	3. REPOSITORY (HEAD)
Files I am actively editing on disk. Change here → git add	The proposed snapshot for my next commit. git add → staged snapshot	Committed snapshots stored in .git . git commit → history
Typical flow: edit files → git status → git add → git diff --staged → git commit → git log		

References — Git Documentation: git-scm.com/docs and Git Book: git-scm.com/book/en/v2 (accessed Sept. 17, 2026); GNU Coreutils Manual: gnu.org/software/coreutils/manual/ (accessed Sept. 17, 2026). Descriptions and examples are paraphrased.